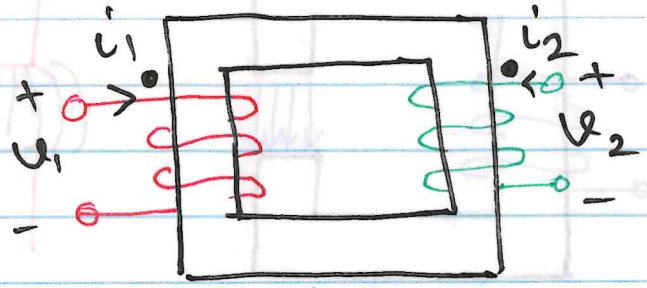
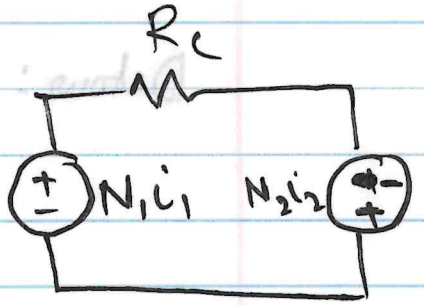
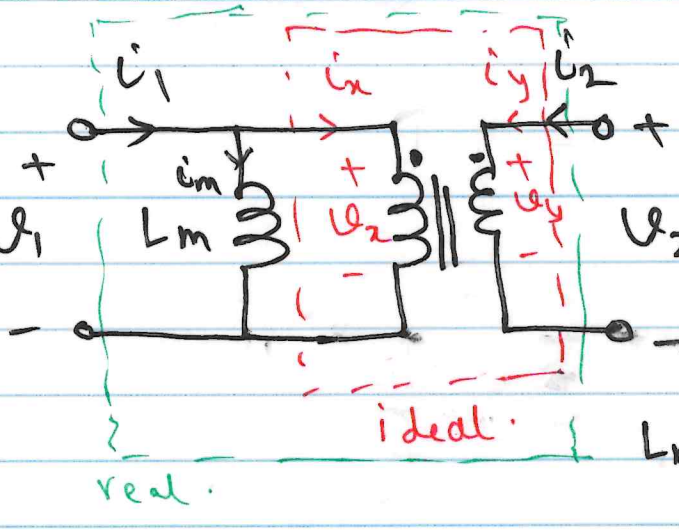
Lecture 10







$$\frac{v_x}{N_1} = \frac{v_y}{N_2}$$

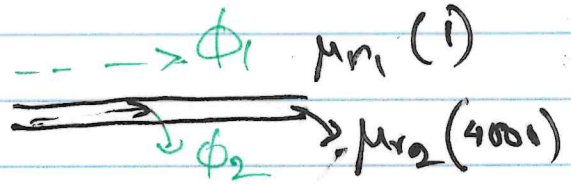
$$N_1 i_x + N_2 i_y = 0$$

$$i_1 = i_m + i_x$$

$$L_m \frac{d i_m}{dt} = v_1$$

Finite permeability.

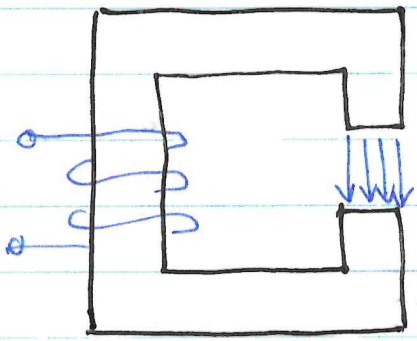
$R_c \equiv$ finite $\quad L_m = \dfrac{N_1^2}{R_c}$

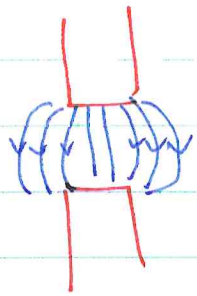


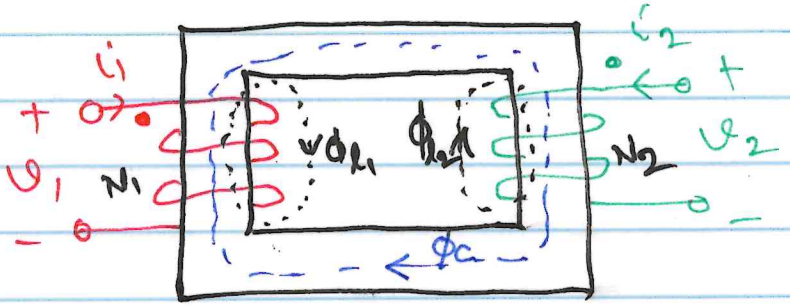
$\phi_2 \gg \phi_1$

$\mu_{r_2} \gg \mu_{r_1}$

Detour:

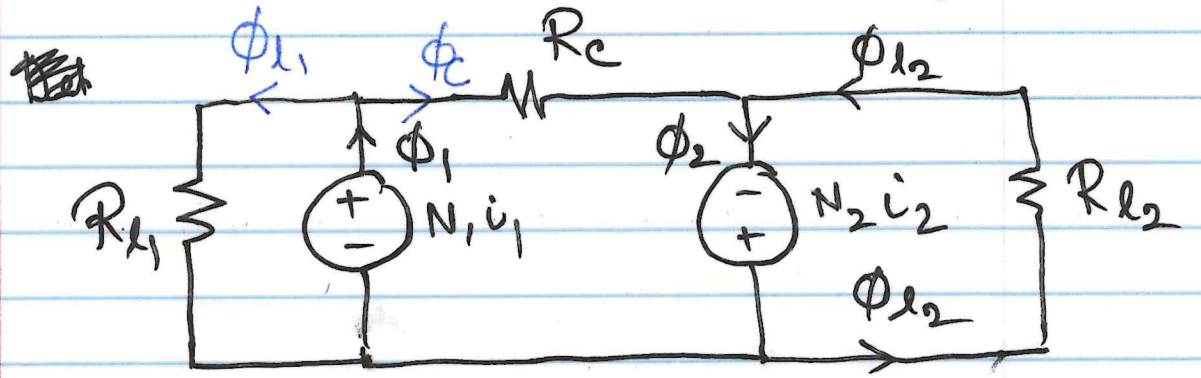






ϕ_{l_1}, ϕ_{l_2} = leakage flux

ϕ_c = Mutual flux, flux coupled by both coils.



$$\phi_1 = \phi_c + \phi_{l_1}$$

$$\phi_2 = \phi_c + \phi_{l_2}$$

$$\phi_{l_1} = \frac{N_1 i_1}{R_{l_1}} \qquad \phi_{l_2} = \frac{N_2 i_2}{R_{l_2}}$$

$$\phi_c = \frac{N_1 i_1 + N_2 i_2}{R_c}$$

$$\phi_1 = \frac{N_1 i_1}{R_{\ell_1}} + \frac{N_1 i_1 + N_2 i_2}{R_c}$$

$$= \left(\frac{N_1}{R_{\ell_1}} + \frac{N_1}{R_c}\right) i_1 + \frac{N_2}{R_c} \cdot i_2$$

$$\boxed{v_1 = N_1 \frac{d\phi_1}{dt}}$$

$$\boxed{v_1 = \left(\frac{N_1^2}{R_{\ell_1}} + \frac{N_1^2}{R_c}\right)\frac{d}{dt} i_1 + \frac{N_1 N_2}{R_c}\frac{d}{dt} i_2}$$

L_{11} : self inductance of #1

M : Mutual inductance

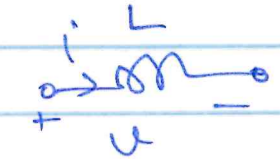
$$v_2 = \left(\frac{N_2^2}{R_{\ell_2}} + \frac{N_2^2}{R_c}\right)\frac{d}{dt} i_2 + \frac{N_1 N_2}{R_c}\cdot\frac{d}{dt} i_1$$

L_{22} : self inductance of #2

$$v_1 = L_{11}\frac{di_1}{dt} + M\frac{di_2}{dt}$$

$$v_2 = M\frac{di_1}{dt} + L_{22}\frac{di_2}{dt}$$

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} L_{11} & M \\ M & L_{22} \end{bmatrix} \frac{d}{dt}\begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$$



$$v = L\frac{di}{dt} \qquad \bar{V} = j\omega L \bar{I}$$

$$\bar{V}_1 = j\omega L_{11}\bar{I}_1 + j\omega M \bar{I}_2$$

$$\bar{V}_2 = j\omega M \bar{I}_1 + j\omega L_{22}\bar{I}_2$$

Coupling coefficient : $K = \dfrac{M}{\sqrt{L_{11}L_{22}}}$ $\qquad 0 \le K \le 1$

$$\frac{v_x}{N_1} = \frac{v_y}{N_2} \qquad N_1 i_x + N_2 i_y = 0$$

$$v_y = v_2 \qquad i_y = i_2$$

$$i_1 = 0$$

$$i_x = -\frac{N_2 i_2}{N_1}$$

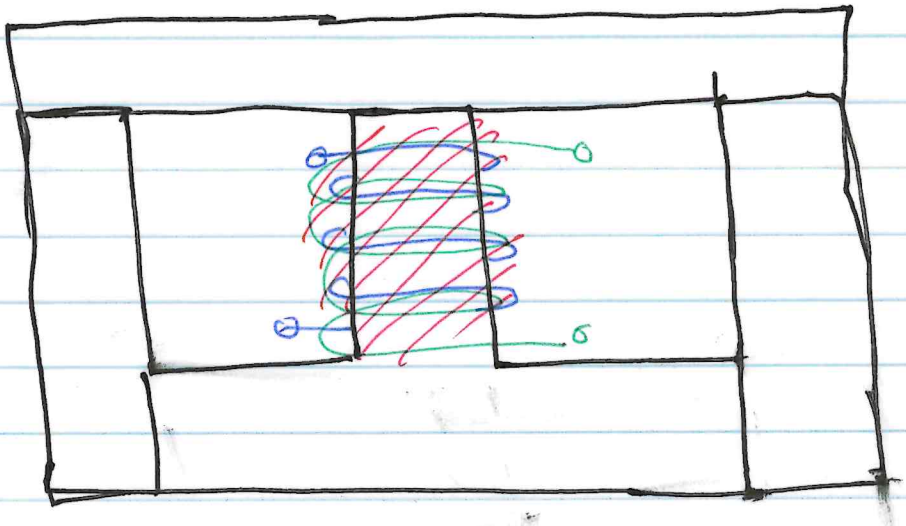
$$v_x = \frac{N_1}{N_2} \cdot v_y = \frac{N_1}{N_2} \cdot v_2$$

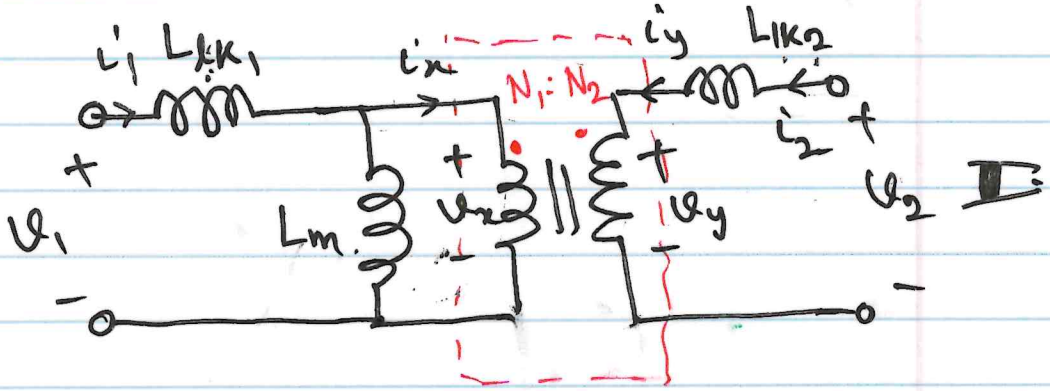
$$\boxed{v_x = -L_m \frac{di_x}{dt}}$$

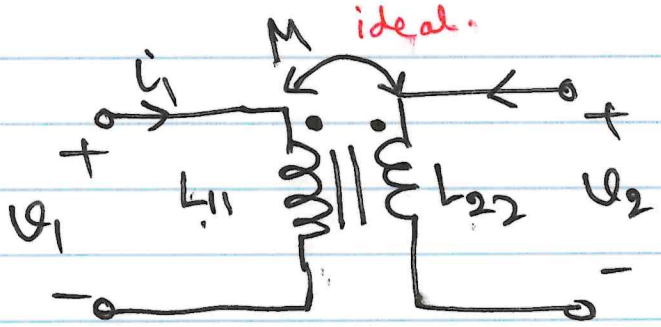
$$\frac{N_1}{N_2} v_2 = -L_m \frac{d}{dt}\left[-\frac{N_2}{N_1} \cdot i_2\right]$$

wdg #1 oc No leakage.

$$\boxed{v_2 = L_m \left(\frac{N_2}{N_1}\right)^2 \frac{d}{dt} i_2}$$







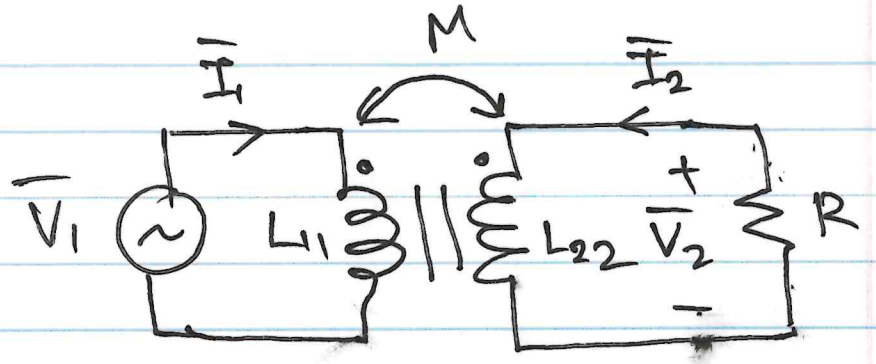
Wdg #1 open circuit ; Wdg #2

$i_1 = 0$

$$v_2 = \boxed{L_{eq}} \frac{di_2}{dt}$$

no leakage

$$\left[\begin{array}{l} L_{1K_1} = 0 \\ L_{1K_2} = 0 \end{array} \right]$$



$$\bar{V}_1 = j\omega L_{11}\bar{I}_1 + j\omega M\bar{I}_2$$

$$\bar{V}_2 = j\omega M\bar{I}_1 + j\omega L_{22}\bar{I}_2$$

$$\bar{V}_2 = -\bar{I}_2 R$$